

Course Outline
CSE 321: Operating Systems
Semester: Summer 2026

Course Description:

This course provides a general understanding of the principles and concepts governing the functions of operating systems and acquaints students with the state-of-the-art approaches that make the design, implementation, and operation of complex operating systems possible. The core of the course contains concurrent programming (threads and synchronization), inter-process communication, process and thread management, storage and resource management and communication, and an analysis on security and protection.

Prerequisites: CSE221: Algorithms

Co-requisites: None

Course Outcomes:

CO 1	Understand the concepts of processes and their management, CPU scheduling techniques, and Apply system calls for solving problems on process management, and Analyze security concerns, and protection mechanisms of OS..
CO 2	Understand the behaviors of threads and their management, process, and thread synchronization mechanisms, deadlocks, virtual machines and containers, Apply thread libraries for solving problems on thread management and synchronization, and Understand mechanisms of file systems concepts, and their implementations in OS, and Apply the concepts to design complex file systems.
CO 3	Understand mechanisms of main memory, and virtual memory management in OS.

Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	PO	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO 1	Understand the concepts of processes and their management, CPU scheduling techniques, and Apply system calls for solving problems on process management, and Analyze security, and protection of OS.	PO1	Cognitive/ Apply, Analyze	Lectures, notes, simulation demo, lab class	Midterm Exam, Final Exam, Lab Work
CO 2	Understand the behaviors of threads and their management, process, and thread synchronization mechanisms, deadlocks, virtual machines and containers, Apply thread libraries for solving problems on thread management and synchronization, and Understand mechanisms of file systems concepts, and their implementations in OS, and Apply the concepts to design complex file systems.	PO2	Cognitive/ Apply, Analyze	Lectures, notes, simulation demo, lab class	Midterm Exam, Final Exam, Lab Work
CO 3	Understand mechanisms of main memory, and virtual memory management in OS.	PO5	Cognitive/ Understand, Apply, Analyze	Lectures, notes, simulation demo, lab class	Final Exam, Lab Work

Lesson Plan:

Sr. No.	Topic details	Time allocation	Related CO
1	Process concepts, states, Inter Process Communication (IPC), operation on the process (creation and termination).	Week 1, 2, 3	CO1
2	Threads overview, Multicore Programming, Multithreading Models, Thread Libraries, Threading Issues, and operating systems examples.	Week 3, 4	CO2
3	Basic concepts of CPU scheduling, scheduling criteria, scheduling algorithms and simulations (FCFS, SJF, Priority, Round Robin), and Multilevel queue and Multilevel feedback queue algorithms.	Week 4, 5	CO1
4	Concepts of Process Synchronization, Race Condition, Critical Section Problem, Mechanisms of Process Synchronization: Test and Set, Compare and Swap, Mutex locks, Semaphores, and Classical synchronization problems.	Week 5, 6, 7	CO2
5	Magnetic Disk Structure Overview, Magnetic Disk Architecture, File Concepts, File Attributes, File Operations, Indexed File Allocation Method, UNIX Inode, File System Implementation, Crash consistency and journaling.	Week 9, 10, 11	CO2
6	Main Memory Management Background, Address spaces, paging, implementation of page table, shared paging, hierarchical page table. Introduction of virtual memory, demand paging, the performance of demand paging, page fault handling, page replacement algorithms (FIFO, LRU, Optimal).	Week 10, 11, 12	CO3
7	Security Problem, security violation categories and methods, security measure levels, program threats, implementing security defenses, firewall, goals and principles of protection, domain of protection, domain structure, domain implementation in unix and android, access matrix, revocation of access rights, mandatory access control, capability-based systems, other protection improvement methods and language-based protection.	Week 12, 13	CO1
8	Concepts of Virtual Machines and Containers and differences between them, Concepts of Hypervisors, Mobile OS concepts.	Week 13, 14	CO2

Lab Plan:

Class	Week	Time Allocation	Topic	Description	Objectives
0	1-2	9 June - 18 June	Basic Shell Commands, Basic C Programming	Online tutorial(s) provided in the buX of basic Linux shell/bash commands and the fundamentals of the C programming language.	Familiarization of students with basic shell commands, and C programming language to orient them with future lab activities.
1	3	20 June - 25 June	System Calls	Lecture on system calls in C: process manipulation, file/file descriptor manipulation, performance profiling, system call tracing.	Create the fundamentals of system calls for students to apply them in xv6 and further labs.
2	4	27 June - 2 July	Linux Shell Scripting	Lecture on shell/bash scripting	Introduces students to Linux shell scripting fundamentals. Students will learn to automate system tasks, handle user input, and work with command-line arguments.
3	5	4 July - 9 July	Assessment 1	Assessment on basic shell commands and basics of C programming, which were learnt by self-study from the online tutorials of buX.	
4	6	11 July - 16 July	Threads and Synchronization	Lecture on creation and management of pthreads and thread synchronization mechanisms using mutex and semaphore.	Learning how to create and manage multiple threads using pthreads and solving race condition problems within multiple threads using mutex and semaphore.

5	7	18 July - 23 July	IPC	Lecture on IPC mechanisms such as pipes, shared memory, and message passing.	Learning and implementing various techniques of Inter-Process Communications such as pipes, shared memory and message passing in order to establish communication among multiple processes.
	8, 9	25 July - 6 Aug	Mid-Week - No Lab		
6	10	8 Aug - 13 Aug	xv6 - Introduction	Introductory lecture on xv6: introduction, xv6 installation tutorial, booting process (?), kernel vs user space, file system hierarchy.	Teach students what xv6 is, how to run it on their own devices, explain the xv6 codebase, create user programs.
7	11	15 Aug - 20 Aug	xv6 - Features	In-depth explanation: system call implementation and tracing, process structure, context switching, scheduler control.	Build on syscall fundamentals to teach implementation in xv6, teach how preemptive process scheduling works in xv6.
8	12	22 Aug - 27 Aug	Assessment 2	Assessment on system calls and threads and synchronization which were covered in lab classes 1 and 4.	
9	13	29 Aug - 3 Sep	Review & Project Viva		

Course Assessment Methods:**Homework**

Homework/assignments shall be designed to ensure that the students have the required knowledge to analyze and design control systems. Specifically, they will support the student's progress in the project/Laboratory.

Quizzes

Quizzes will be designed to test the student's understanding of the course and to assess various course outcomes

Examinations

The exam shall contain problems designed to test knowledge and comprehension, to analyze control systems, and/or apply the engineering problem-solving method.

Laboratory Work

The students will have hands-on experience in the design and analysis of shell commands and scripts in Linux OS. They will learn how to work with different tools of the OS such as system calls, pthreads, synchronization techniques and implement schedulers at the OS level. The students will be assessed by multiple lab assignments, lab exams and a term-project.

Assessment Methods vs. Course Outcomes:

Assessment Methods	CO 1	CO 2	CO 3
Midterm Exam	X	X	X
Final Exam	X	X	X
Laboratory Works	X	X	X

Textbook:

- 1 Operating System Concepts Essentials by Abraham Silberschatz, Peter B. Galvin and Greg Agne, Wiley; 10th Edition (July 5, 2008). ISBN: 978-1119320913
- 2 Modern Operating Systems by Tanenbaum A.S., Prentice Hall; 4th Edition (2015). ISBN-13: 978-0133591620
- 3 Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Arpaci-Dusseau Books, October, 2023 (Version 1.10), <http://www.ostep.org>

Course Policies:**Class Policy**

- Classroom and laboratory attendance are mandatory. You should come to the classroom before the instructor. Latecomers may/ may not be allowed to enter the classroom. Students, who are absent over 30% of the class time will not be allowed to enter the final examination.
- You should turn off your cellular phone before entering the classroom. You should not leave the classroom to make or take cellular phone calls.
- You should bring a notepad and/or a writing instrument to every class and take

detailed notes.

- You should pay attention to the instructor and participate in class discussions.
- You should not do other work during class time.

Honor Code

Any form of cheating, plagiarism, and/or academic dishonesty will result in an "F" grade in the course.

Late Work and Examinations

Late assignments will not be accepted. Students who know that they are going to miss class should make arrangements in advance. Exams will be closed book. There will not be any make-up for quizzes and midterm exams except the cases of hospitalization or detention.

Grading Policies:

Students' grades are assigned according to the grading scale of the Brac University Undergraduate Study and Examinations Regulations. In addition, the faculty are allowed to take into consideration the class average and standard deviation to reflect the actual class performance for student grade assignments. The grades at the university will be indicated in the following manner:

Marks	Grades
90-100	A (4.0)
85- <90	A- (3.7)
80- <85	B+ (3.3)
75- <80	B (3.0)
70- <75	B- (2.7)
65- <70	C+ (2.3)
60- <65	C (2.0)
57- <60	C- (1.7)
55- <57	D+ (1.3)
52- <55	D (1.0)
50- <52	D- (0.7)
<50	F (0.0)
P	Pass
I	Incomplete
W	Withdrawal
R	Retaken

Course Assessment Methods:

Guidelines for CSE course teaching in BRAC University. The following assessment methods are based on a Theory Course with Lab.

Section	Marks (%)
1. Assignments	5 %
2. Quizzes/Class Tests	10 %
3. Midterm Examination	25 %
4. Lab & Projects	25 %
5. Final	35 %
Total	100 %

CO-wise tentative marks distribution:

CO	Marks (%)
CO1	30%
CO2	35%
CO3	35%
Total	100 %

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